

Benjamin DeJong

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Professional Summary

Computer Scientist and Mathematician with research experience in AI, LLMs, HPC, and compression. 5 years building performant software in Python, C++ 23, Rust. 3 years working with machine learning frameworks. 5 years working with generative AI for software engineering. Proficient at statistical and mathematical methods to solve complex, real-world problems. Clear communicator, experienced with cross-functional collaboration and technical tutoring. ICPC, Putnam, and ACCA calculus competitor.

Technical Skills

Programming: C++ (5 years), C++23, Python (5 years), CUDA, Rust (4 years), Matplotlib, Git, Docker, Seaborn

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, model evaluation, backtesting, Gen AI, vector embeddings

Systems: Linux (2 years), HPC Clusters, distributed computing, MPICH

Math: Probability, Predictive Modeling, Real Analysis, Differential Equations, Data Visualization, Numerical Methods

Languages: English, Spanish (7 years).

Education

Bachelor of Science at Elmhurst University

(Senior) Anticipated: May 2026

Majors: Mathematics, Computer Science

GPA: 3.84/4.00

Transfer Excellence Scholarship, Honors Program Member

ICPC, Putnam, and ACCA calculus competitor.

Work Experience

Researcher, Argonne National Lab

(4 months) May 2025 - Aug 2025

- Benchmarked compressing collective communication libraries on HPC systems across 128 multi-GPU compute nodes.
- Built Python data pipelines to parse outputs, analyze performance, and visualize scaling behavior, creating insights.
- Identified and explained unexpected behavior through empirical scaling tests, targeting problems.
- Fixed flawed benchmark code by modifying C++/CUDA timing code, enabling accurate millisecond-level measurements.

Tutor, Elmhurst University

(8 months) August 2025 - Present

- Explained Calculus I-III, statistics, and algebra rigorously and intuitively to 40+ students.
- Managed tutoring schedules and maintained detailed student progress reports.

LLM Researcher, Elmhurst University

(1 year) January 2024 - December 2024

- Scraped and cleaned datasets to fine-tune LLMs, improving accuracy and reliability.
- Designed and evaluated prompts to improve usefulness, reliability, and mitigate prompt-injection attacks.
- Collaborated with a cross-functional team to give an accurate, documented, reliable product to an external company.

Tech Support & Sales Associate, Perry's Music

(6 years) November 2018 - May 2025

- Assisted 1000s of customers, negotiated 100s of sales online and in person in written and spoken English and Spanish.
- Configured devices and solved technological computer problems.
- Managed a 200 item inventory using Microsoft Excel and created online item listings, enabling targeted advertisement.

Projects

Blackjack Models (2 years)

- Applied Deep Q-Learning and classical algorithms to learn near-optimal policies, achieving a 45% win rate.
- Implemented a Blackjack environment from scratch as a Markov Decision Process in Python.
- Evaluated policy performance through large-scale simulation and statistical analysis.

Market Microstructure and Equilibrium Modeling (2 years)

- Modeled realistic market matching dynamics by building a limit order book simulator with priority queues in Python.
- Analyzed emergent price formation through large-scale stochastic trade simulations using numerical methods.
- Learned market equilibrium prices with a neural regression model (PyTorch) trained on simulated markets.

- Captured buyer-seller interactions with nonlinear differential equations solved using RK45 and Euler methods.
- Utilized XGBoost, BERT for financial news analysis to predict stock movements during market closed periods.

Custom Programming Language Compiler (1 year)

- Developed a formal grammar for a statically-typed imperative language.
- Classified identifiers, numeric literals, operators, and reserved keywords using a custom lexer implemented with regex.
- Validated syntax and expressions using a custom recursive-descent parser operating on a token stream.
- Applied compiler front-end techniques including token lookahead, grammar-driven parsing, and error detection.

Car Price Predictor (2 years)

- Built an end-to-end data pipeline, including web scraping, cleaning, and preprocessing with BS4 and Pandas.
- Performed feature engineering by tokenizing variables for supervised regression.
- Achieved 95% accuracy utilizing scikit-learn decision trees.
- Compared PyTorch neural networks against classical models to analyze tradeoffs and complexity.